

Arunabha Majumder

Mechanical Designer — 3D Design, Strength Calculation (FEA), Drawings & Prototyping

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October 7, 2026

Hiring Team
Hefmec
Vantaa, Finland

Application: Tool Designer (Työvälinesuunnittelija)

Dear Hiring Team,

I am applying for the Tool Designer position at Hefmec. I am a mechanical engineer with a BTech in Mechanical Engineering and an MTech in Mechatronics, finishing my PhD at Aalborg University, with about three years of hands-on 3D mechanical design and strength calculation. What attracts me to Hefmec is that your designs do not stay on the screen: you follow each tool from idea to commissioning and make sure it works in the customer's production.

In my PhD, I took ownership of the design of an actuated shoulder exoskeleton from concept to tested hardware. I carried out the concept design, 3D modelling and detail design in SolidWorks, prepared workshop drawings for machined and 3D-printed parts, and used strength calculation with FEA to check the stiffness, strength and weight of the load-carrying parts before manufacture. I then assembled and tested the prototypes myself and improved the design based on the results. The design reduced energy consumption by more than 25% and won the Best Paper Award at IFToMM ISRM 2026.

I solve technical problems in a practical way. On a 5-bar planar parallel robot, I developed a method to estimate the end-effector force from the actuators and validated it against a force sensor to within 10% error. During my Master's at CSIR-CMERI, I built pneumatic actuators from raw materials and set up the test rig myself.

I have always built things with my hands. As a student, I designed and built a Go-Kart from the ground up and helped build a human-electric hybrid vehicle for the Efficycle competition. In every project since, I have made the parts, assembled them and fixed what did not work.

I want to be open about what I would learn with you. I have not yet prepared CE documentation or worked formally with machine safety standards, and I currently work in English rather than Finnish. These are exactly the skills I want to build, and Hefmec's step-by-step introduction alongside an experienced designer is the right setting for me to grow towards independent project responsibility.

I would welcome the opportunity to discuss how I can contribute to your team. References are available upon request.

Kind regards,
Arunabha Majumder